ENRICO LAOH, Ph.D.

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https://elaoh.github.io | https://www.linkedin.com/in/laohenrico

Brief About Me

- I am an **interdisciplinary researcher with a PhD in Industrial Engineering and Management**, working at the **intersection of artificial intelligence, information technology, life science analytics,** and **managerial aspects,** with a focus on developing **adaptive, interpretable, and human-centered intelligent systems to support decision-making in complex systems**. My research advances **AI-driven decision frameworks** by integrating **machine learning, explainable AI, and optimization** to enable reliable, transparent, and adaptive decision-making.
- My **current** research spans **healthcare and biomedical analytics, genomics analytics, environmental analytics, and intelligent decision support systems**, emphasizing the translation of complex multimodal data into **actionable insights for sustainable and evidence-based decisions**.
- Supported by a strong record of **peer-reviewed publications, international awards, competitive fellowships, and leadership experience**, including guiding an **INFORMS student chapter to national recognition**, I am committed to advancing **interdisciplinary and translational AI research** with real-world impact and commercialization potential, reinforced by recognition as a **NASA-funded I-Corps Creativity, Innovation, and Entrepreneurship Scholar**.

Educations

Doctor of Philosophy, OKLAHOMA STATE UNIVERSITY — July 2026
Focus Area/Major: Industrial Engineering and Management
Degree GPA: 4.00/4.00

Graduate Certificate, OKLAHOMA STATE UNIVERSITY — July 2023
Focus Area/Major: Business Analytics and Data Science
Degree GPA: 4.00/4.00

Master of Science, OKLAHOMA STATE UNIVERSITY — December 2022
Focus Area/Major: Industrial Engineering and Management
Degree GPA: 4.00/4.00

Master of Engineering, UNIVERSITAS INDONESIA — July 2017
Focus Area/Major: Data Engineering and Quality
Degree GPA: 4.00/4.00

Bachelor of Engineering, UNIVERSITAS INDONESIA — July 2016
Focus Area/ Major: Industrial Engineering
Degree GPA: 3.88/4.00

Certifications: *Tableau Desktop Specialist*, *Amazon SageMaker Studio*, *Amazon Redshift*, *AWS MLOps*, *Six Sigma Green Belt*

Selected Awards and Honors

- **Graduate Student Leadership Award**, School of Industrial Engineering and Management, 2025
  Recognizes a graduate student who has demonstrated outstanding leadership abilities in student organizations.
- **Lisa Zaken Award for Excellence**, Institute of Industrial and Systems Engineers, 2024
  This international annual award, given only to one winner who has outstanding scholastic achievement, leadership, and dedication to the Industrial and Systems Engineering profession, demonstrates the highest of standards.
- **Third place in the 2024 IISE ESD/QCRE/PG&E Energy Analytics Challenge**, 2024
  Annual international competition by the Energy Analytics Society, judged by academia and industry practitioners.
- **President Leadership Recognition**, Hargis Leadership Institute, 2024
  Awarded for demonstrating exemplary leadership and impact within the university and broader community.
- **Outstanding Graduate Student Award**, CEAT, Oklahoma State University, 2024
  Selected as the top graduate student in the CEAT based on academic excellence and research achievements.
- **Robberson Dissertation Fellowship**, Oklahoma State University, 2024
  Recognized for outstanding doctoral research contributing to advancements in industrial engineering and data science.
- **Roy and Virginia Dorrough Distinguished Graduate Fellowship**, Oklahoma State University, 2023, 2024, 2025, 2026
  Awarded to recognize outstanding graduate students with evidence of exceptional academic performance and achievements, leadership experiences, and notable extracurricular or community engagement activity.
- **Creativity, Innovation, and Entrepreneurship Scholar**, Spears School of Business, Oklahoma State University, 2021
  Distinguished student with excellence, innovativeness, creativity, responsibility for change, engagement in one's surroundings, personal integrity, and strong moral character.
- **IEEE SMC Society Merit Paper Awards**, 8th International Conference on Awareness Science and Technology (iCAST), 2017
  Given to the most outstanding paper in the IEEE SMC academic society annually that is presented at the conference.
- **IEEE Indonesia Section Best Presenter Award**, 3rd International Conference on Science in Information Technology (ICSITech), 2017
  Given to the best-presented paper at the conference.

Work Experiences

COMPUTATIONAL GENOMIC RESEARCH SCIENTIST — January 2026 to **present**
TRANSLATIONAL GENOMICS LABORATORY, OKLAHOMA STATE UNIVERSITY, USA
Collaborate with multidisciplinary omics teams to integrate multi-omics data and study the pathway from DNA to RNA to proteins and phenotypes. Apply computational modeling to convert large-scale biological data into testable hypotheses, leveraging expanding genomic resources. Advance scalable, whole-genome, and spatially adaptive approaches to improve phenotype assessment and variety development.

RESEARCH ASSOCIATE — January 2026 to **present**
FACULTY OF AGRICULTURAL LIFE & ENVIRONMENTAL SCIENCES, UNIVERSITY OF ALBERTA, CANADA
Analyze large-scale genomic and phenotypic data to investigate genetic variation,

population structure, and trait associations in forest tree species, particularly lodgepole pine and white spruce. Apply Ancestral Recombination Graphs (ARGs), GWAS, gene annotation, and evolutionary analyses to identify genetic markers and biological mechanisms associated with forest growth and resilience. Develop a genomically informed Multi-Agent Reinforcement Learning (MARL) framework to support adaptive forest management, integrating genetic insights, ecological sustainability, and economic objectives.

DATA SCIENCE EXPERT — August 2019 to **present**

INDUSTRIAL ENGINEERING VENTURE UNIT, UNIVERSITAS INDONESIA, INDONESIA

Provide optimal consulting and training services by leveraging expertise for private/government institutions in Indonesia.

DATA SCIENCE INTERN — June 2024 to August 2024

UNITED AIRLINES INC., CHICAGO, USA

Observed the pricing main model and understood the effect of demand forecasting. Updated the demand forecasting model and tested novel variables extracted from internal databases and external potential influences. Reduced forecasting *MAPE by 2%.*

HEALTH DATA ANALYTICS RESEARCHER — October 2022 to July 2026

CENTER FOR HEALTH SYSTEMS INNOVATION, OKLAHOMA STATE UNIVERSITY, USA

Developed frameworks and models to analyze healthcare data and synthesize knowledge to support a medical-based decision-making system. *Funded by a $1.2M NIH Grant*.

TEACHING AND RESEARCH ASSISTANT — January 2021 To December 2025

SCHOOL OF INDUSTRIAL ENGINEERING AND MANAGEMENT, OKLAHOMA STATE UNIVERSITY, USA

Taught tutorials and courses in Industrial Engineering, Operations Management, and Data Analytics. Did research on collecting, extracting, and building interpretable and explainable Machine Learning models from human physiological data (EEG, ECG, Eye Motion, etc.) to measure stress levels and cognitive state to help industrial AI-based systems generate robust decisions.

NON-TENURE TEACHING PROFESSOR and RESEARCHER — August 2017 to December 2020

INDUSTRIAL ENGINEERING DEPARTMENT, UNIVERSITAS INDONESIA, INDONESIA

Researched the application of descriptive, predictive, and prescriptive analytics to a wide range of industrial problems, including network optimization, power generation, customer segmentation, natural language processing, and supply chain design. Taught and conducted classes related to industrial engineering and data analytics.

Professional Memberships

- **Institute for Operations Research and the Management Sciences (INFORMS)**
  OSU Chapter President (2022-2023): Won the cum laude chapter award. Member since 2019.
- **Institute of Industrial and Systems Engineers (IISE)**
  Member since 2021.
- **American Medical Informatics Association (AMIA)**
  Member since 2025.
- **Eastern North American Region International Biometric Society (ENAR IBS)**
  Member since 2026.

Research and Publications

Metrics: h-index 13 | i10-index 16 | 500 citations per September 2026

Patents

PT002. "**Permissioned Blockchain System for Halal Supply Chain Through Hyperledger Fabric Web-Based Application**" Indonesian Patent (Submitted)

PT001. "**Product Service Systems for CNC (Computer Numerically Controlled) Lathes Machine Maintenance using Integrated Web-Based Application**" (2024) Indonesian Patent No. IDP000092940

PT0xx = i-th patent

Journal Articles and Conference Proceedings

Healthcare Analytics

J011. Dhini, A., Dwi, K., Surjandari, I., **Laoh, E.**, Dzikri, A., "Prediction of LOS for Heart Disease Patient using Ensemble Machine Learning Approach." (working paper).

J010. Bani Ahmad, O., **Laoh, E.**, Liu, T., "Risk-Confidence Ordinal Scoring (RCOS): A Decision-Optimization and Calibration Framework for Early Stroke Subtype Prediction from Routine Electronic Health Records." (Computers in Biology and Medicine - submitted).

J009. **Laoh, E.**, Liu, T., "Incremental Learning with Multi-Group Data Stream for Diabetic Retinopathy Prediction" (IEEE Transactions on Knowledge and Data Engineering – under review).

J008. **Laoh, E.**, Adelia, F., Zulkarnain, "Temporal Neuron based Artificial Neural Network Model for Blood Components Demand Forecasting" (Journal of Medical Internet Research Artificial Intelligence – under review).

J005. Bani Ahmad, O., **Laoh, E.**, Liu, T., "Temporal Machine Learning Models for Early Stroke Subtype Prediction Using Laboratory Data." (2025) International Journal of Medical Informatics – preprint, DOI: 10.2139/ssrn.5680004

J004. **Laoh, E.**, Bani Ahmad, O., Liu, T., "A robust and trustable approach to incorporate medical domain knowledge in machine learning models for diabetic retinopathy screening using routine lab results" (2026) Smart Health Journal, Volume 42, 2026, 100724, DOI: 10.1016/j.smhl.2026.100724.

C023. Bani Ahmad, O., **Laoh, E.**, Liu, T., "Designing a Pruning and Merging Method to Achieve Simple Rules for Diabetic Retinopathy Screening with Routine Lab Results." (2025) Institute of Industrial and Systems Engineers Annual Proceedings, DOI: 10.21872/2025IISE_8832

C022. Rahmadianti, R., Dhini, A., **Laoh, E.**, "Estimating customer lifetime value using LRFM model in pharmaceutical and medical device distribution company." (2020) International Conference on ICT for Smart Society (ICISS), DOI: 10.1109/ICISS50791.2020.9307592

C014. Pramono, P.P., Surjandari, I., **Laoh, E.**, "Estimating customer segmentation based on customer lifetime value using two-stage clustering method." (2019) International Conference on Service Systems and Service Management, DOI: 10.1109/ICSSSM.2019.8887704

Behavioral Analytics and Text Mining

C021. Dhini, A., Budiani, L.R., **Laoh, E.**, "Segmenting and Targeting the Potential Markets of a Muslim Fashion Company." International Conference on ICT for Smart Society (ICISS) 2020, DOI: 10.1109/ICISS50791.2020.9307604.

J001. Surjandari, I., Wayasti, R.A., **Laoh, E.**, Zulkarnain, Rus, A.M.M., Prawiradinata, I. "Mining public opinion on ride-hailing service providers using aspect-based sentiment analysis." International Journal of Technology, Vol 10, No 4 (2019) pages: 818-828. DOI: 10.14716/ijtech.v10i4.2860.

C013. Prabaningtyas, N.I., Surjandari, I., **Laoh, E.**, "Mining customers opinion on services and applications of mobile payment companies in Indonesia using sentiment analysis approach." Proceeding of 16th International Conference on Service Systems and Service Management, ICSSSM, Shenzhen: 2019, DOI: 10.1109/ICSSSM.2019.8887643.

C011. **Laoh, E.**, Surjandari, I., Febirautami, L. R. "Indonesian's Song Lyrics Topic Modelling using Latent Dirichlet Allocation." Proceeding of 5th International Conference on Information Science and Control Engineering, Zhengzhou: 2018, pp. 270-274. DOI: 10.1109/ICISCE.2018.00064.

C010. Febirautami, L. R., Surjandari, I., **Laoh, E.** "Determining Characteristics of Popular Local Songs in Indonesia's Music Market." Proceeding of 5th International Conference on Information Science and Control Engineering, Zhengzhou: 2018, pp. 197-201. DOI: 10.1109/ICISCE.2018.00050.

C009. Azitha, R., Surjandari, I., **Laoh, E.** "Mining Information Search Pattern on Website: A case study of educational institution." Proceeding of 5th International Conference on Information Science and Control Engineering, Zhengzhou: 2018, pp. 326-330. DOI: 10.1109/ICISCE.2018.00075.

C008. Surjandari, I., Rosyidah, A., Zulkarnain, **Laoh, E.** "Mining Web Log Data for News Topic Modeling Using Latent Dirichlet Allocation." Proceeding of 5th International Conference on Information Science and Control Engineering, Zhengzhou: 2018, pp. 331-335. DOI: 10.1109/ICISCE.2018.00076.

C007. Prameswari, P., Surjandari, I., **Laoh, E.** "Opinion mining from online reviews in Bali tourist area," 2017 3rd International Conference on Science in Information Technology (ICSITech), Bandung, 2017, pp. 226-230. DOI: 10.1109/ICSITech.2017.8257115.

C006. Prameswari, P., Zulkarnain, Surjandari, I., **Laoh, E.** "Mining online reviews in Indonesia's priority tourist destinations using sentiment analysis and text summarization approach," 2017 IEEE 8th International Conference on Awareness Science and Technology (iCAST), Taichung, Taiwan, 2017, pp. 121-126. DOI: 10.1109/ICAwST.2017.8256429.

Optimization and Operations Management

J007. Khrisnan, D., R., **Laoh, E.**, Liu, T., "Multi vehicle pickup-and-delivery problem with time windows and handling costs." (Transportation Research Part B - under review).

J006. Puspitadewi, C. H., Arian Dhini, A., **Laoh, E.**, Sudiana, D., "GIS-based wildfire prediction model in Indonesia using stacking ensemble learning." Artificial Intelligence in Geosciences, Volume 7, Issue 2, 2026, DOI: 10.1016/j.aiig.2026.100228.

J003. Surjandari, I., Dzikri, A., Dhini, A., **Laoh, E.**, Dwi, K., Ferrouzi, D., "Stacked Generalization with Sequential-Model Based Optimization for estimating Used Car Valuation in Indonesia." (2024) Engineering, Technology & Applied Science Research. DOI: 10.48084/etasr.8226.

J002. Surjandari, I., Yusuf, H., **Laoh, E.**, Maulida, R. "Designing a Permissioned Blockchain Network for the Halal Industry Using Hyperledger Fabric with Multiple Channels and the Raft Consensus Mechanism." Journal of Big Data (2021). Springer. DOI: 10.1186/s40537-020-00405-7.

C020. Syaputra, A., **Laoh, E.**, "Customer Segmentation on Returned Product Customers Using Time Series Clustering Analysis." (2020) International Conference on ICT for Smart Society, DOI: 10.1109/ICISS50791.2020.9307575

C019. Ramadhani, S., Dhini, A., **Laoh, E.**, "Airline Passenger Forecasting using ARIMA and Artificial Neural Networks Approaches." International Conference on ICT for Smart Society (ICISS) 2020, DOI: 10.1109/ICISS50791.2020.9307571

C018. Amira, R.F., Surjandari, I., **Laoh, E.**, "Jakarta Flood Risk Mapping Using Index-based Approach and Spatial Analysis." International Conference on ICT for Smart Society (ICISS) 2020, DOI: 10.1109/ICISS50791.2020.9307583

C017. Ilham, M.Y., Surjandari, I., **Laoh, E.**, "Analyzing highway road accident characteristic using data mining." International Workshop on Big Data and Information Security, IWBIS 2020, DOI: 10.1109/IWBIS50925. 2020.9255655.

C016. Hidayattullah, S., Surjandari, I., **Laoh, E.**, "Financial statement fraud detection in Indonesia listed companies using machine learning based on meta-heuristic optimization." International Workshop on Big Data and Information Security, IWBIS 2020, DOI: 10.1109/IWBIS50925.2020.9255426.

C015. Nur Prasasti, I.M., Dhini, A., **Laoh, E.**, "Automobile Insurance Fraud Detection using Supervised Classifiers." International Workshop on Big Data and Information Security, IWBIS 2020, DOI: 10.1109/IWBIS50925. 2020.9255563.

C012. Nadinta, D.S., Surjandari, I., **Laoh, E.**, "A clustering-based approach for reorganizing bus route on bus rapid transit system." Proceeding of 16th International Conference on Service Systems and Service Management, ICSSSM, Shenzhen: 2019, DOI: 10.1109/ICSSSM.2019.8887689.

C005. **Laoh, E.** Surjandari, I., Zulkarnain, "Reconfiguring oil distribution route using graph partitioning and graph optimization," 2017 IEEE 8th International Conference on Awareness Science and Technology (iCAST), Taichung, Taiwan, 2017, pp. 103-108. DOI: 10.1109/ICAwST.2017.8256426.

C004. **Laoh, E.** Surjandari, I., Dhini, A. "Electricity distribution clustering and configuration study using KM-MST," 2017 3rd International Conference on Science in Information Technology (ICSITech), Bandung, 2017, pp. 138-143. DOI: 10.1109/ICSITech.2017.8257099.

C003. Yuniarti, T., Surjandari, I., Muslim, E., **Laoh, E.** "Data mining approach for short term load forecasting by combining wavelet transform and group method of data handling (WGMDH)," 2017 3rd International Conference on Science in Information Technology (ICSITech), Bandung, 2017, pp. 53-58. DOI: 10.1109/ICSITech.2017.8257085.

C002. **Laoh, E.** Surjandari, I., Dhini, A. "Clustered Electricity Distribution Configuration Using KM-MST Algorithm." Proceeding of International Conference on Computer Science and Technology, Seoul: 2016.

C001. Maharani, A., Surjandari, I., Rachman, A., **Laoh, E.** "Graph Theory Approach to Analyze Reliability Index of Gas and Steam Power Plant." Proceeding of International Conference on Computer Science and Technology, Seoul: 2016.

J0xx = i-th journal article; C0xx = i-th conference proceeding
In total, 28 published articles can be found on Google Scholar

Talks and Presentations

P014. " Risk-Confidence Ordinal Scoring (RCOS): A Decision-Optimization and Calibration Framework for Early Stroke Subtype Prediction from Routine Electronic Health Records" IISE Annual Meeting 2026, Arlington, Texas, *May 2026*.

P013. "Temporal Machine Learning Models for Early Stroke Subtype Prediction Using Laboratory Data" IISE Annual Meeting 2026, Arlington, Texas, *May 2026*.

IT004. "Introduction to Data Visualization and Storytelling." Guest Lecture, Department of Electrical and Computer Engineering, Universitas Indonesia, Depok, December 2025.

P012. "Incremental Learning with Multi-Group Data Stream for Diabetic Retinopathy Prediction" INFORMS Annual Meeting 2025, Atlanta, Georgia, *October 2025*.

P011. "An Approach for Electricity Price Predictions: Leveraging Machine Learning Model and Multivariate Time Series Analysis" IISE Annual Meeting 2024, Montreal, Canada, *May 2024*.

IT003. "Big data analytics for healthcare." National Board of Physicians of Indonesia, September 2023.

P010. "A Modified Feature Selection Method to Build Interpretable and Trustable Machine Learning Model." INFORMS Annual Meeting 2023, Phoenix, Arizona, October 2023.

P009. "Human-Centred Explainable Artificial Intelligent (XAI)." INFORMS Annual Meeting 2021, Anaheim, California, October 2021.

IT002. "Data Science and Its Applications." SPWI Seminar Series, Depok, April 2019.

IT001. "Data Mining with R." Data Mining Training SQE UI, Depok, September 2018**.**

P008. "Enhancing hospitality sentiment reviews analysis performance using SVM N-grams method." 16th International Conference on Service Systems and Service Management, Shenzhen, July 2019.

P007. "Indonesian's Song Lyrics Topic Modelling using Latent Dirichlet Allocation." 5th IEEE International Conference on Information Science and Control Engineering, Zhengzhou, July 2018.

P006. "Reconfiguring oil distribution route using graph partitioning and graph optimization.", IEEE 8th International Conference on Awareness Science and Technology, Taichung, Taiwan, November 2017.

P005. "Neural network-based system for detecting and diagnosing faults in steam turbine of thermal power plant.", IEEE 8th International Conference on Awareness Science and Technology, Taichung, Taiwan, November 2017.

P004. "Electricity distribution clustering and configuration study using KM-MST", IEEE 3rd International Conference on Science and Technology, Bandung, Indonesia, October 2017.

P003. "Data mining approach for short term load forecasting by combining wavelet transform and group method of data handling (WGMDH)", IEEE 3rd International Conference on Science and Technology, Bandung, Indonesia, October 2017.

P002. "Opinion mining from online reviews in Bali tourist area", IEEE 3rd International Conference on Science and Technology, Bandung, Indonesia, October 2017.

P001. "Clustered Electricity Distribution Configuration Using KM-MST Algorithm.", International Conference on Computer Science and Technology Workshop, Seoul, South Korea, November 2016.

IT0xx = i-th invited talk; P0xx = i-th presentation

Funding History

PR002. Best Future Forests – Advanced Forest Genomics and Integrative Resource Management System. [750K CAD] (2026) Alberta Innovates & Government of Alberta, Canada *(Researcher)*

G012. Empowering Cloud Computing for Non-image-based Diabetic Retinopathy Screening by Designing an EHR-oriented Incremental Learning Framework. [217K USD] (2023-2025) NIH-NEI: NOT-OD-23-070 *(Researcher)*

G011. Harnessing Tensor Information to Improve EHR Data Quality for Accurate Data-driven Screening of Diabetic Retinopathy with Routine Lab Results. [1.2M USD] (2022-2024) NIH-NEI: 5R01EY033861 *(Researcher)*

G010. Application of Supervised Learning Algorithms in the Service Industry. [90M IDR] (2022) PUTI No. NKB-1337/UN2.RST/HKP.05.00/2022 ***(Co-PI)***

G009. Application of Big Data Analytics to Improve the Quality of Risk Analysis. [90M IDR] (2020) PUTI No. NKB-1702/UN2.RST/HKP.05.00/2020 ***(Co-PI)***

G008. Development of Time Series Forecasting Models Using a Data-Driven Approach. [90M IDR] (2020) PUTI No. NKB-1133/UN2.RST/HKP.05.00/2020 ***(Co-PI)***

G007. Escalating Company Performance Using a Data Mining Approach. [90M IDR] (2020) PUTI No. NKB-1070/UN2.RST/HKP.05.00/2020 ***(Co-PI)***

PR001. Framework and Detailed Process Flow for an Asset Management System aligned with Digital Transformation & Solutions in the Electric Power Generation Industry. [150M IDR] (2020) Java-Bali Power Generation Company. ***(Co-PI)***

G006. Application of Data Mining to Accelerate Performance Improvement in the Manufacturing and Service Sectors. [120M IDR] (2019) PIT-9 No. NKB-0061/UN2.R3.1/HKP.05.00/2019 ***(Co-PI)***

G005. Improving Operational Effectiveness and Efficiency Through Data-Driven Method Approaches. [90M IDR] (2018) PITTA No. 2455/UN2.R3.1/HKP.05.00/2018 ***(Co-PI)***

G004. Enhancing the Competitiveness of Indonesia's Service Industry Through Big Data Analysis. [90M IDR] (2018) PITTA No. 787/UN2.R3.1/HKP.05.00/2017 *(Researcher)*

G003. Application of Data-Driven Methods to Improve the Performance of State-Owned Enterprises. [90M IDR] (2017) PITTA No. 788/UN2.R3.1/HKP.05.00/2017 *(Researcher)*

G002. Achieving Electrical Energy Resilience in Indonesia Through Improving Reliability, Forecasting Quality, and Electricity Distribution Using Data Mining and Graph Theory. [90M IDR] (2016) PITTA No. 2132/UN2.R12/HKP.05.00/2016 *(Researcher)*

G001. Economic Indicator Data Analysis Using Data Mining to Predict Company Performance and Financial Fraud in Order to Improve Competitiveness and the Investment Climate in Indonesia. [90M IDR] (2016) PITTA No. 2134/UN2.R12/HKP.05.00/2016 *(Researcher)*

G0xx = i-th government funding; PR0xx = i-th private funding

Teaching

Average instructor rating: 5.33 of 6.00 | > 3 years full-time instructor experience | 5 years teaching assistant

- IEM 4723: Information Systems Design and Development, Oklahoma State University
- IEM 5603: Project Management, Oklahoma State University
- IEM 5990: Advanced Data Analytics, Oklahoma State University
- IEM 4913: Senior Design, Oklahoma State University
- IEM 3503: Engineering Economic Analysis, Oklahoma State University
- IEM 3513: Economic Decision Analysis, Oklahoma State University
- IEM 3813: Work Design, Ergonomics, and Human Performance, Oklahoma State University
- ENGE600001: Calculus, Universitas Indonesia
- ENGE600002: Linear Algebra, Universitas Indonesia
- ENGE600011: Engineering Economics, Universitas Indonesia
- ENIE601002: Introduction to Economics, Universitas Indonesia
- ENIE601004: Probability Theory, Universitas Indonesia
- ENIE603006: Basic Statistics, Universitas Indonesia
- ENIE603007: Linear Programming, Universitas Indonesia
- ENIE604011: Industrial Statistics, Universitas Indonesia
- ENIE604015: Computational Method for IE, Universitas Indonesia
- ENIE604016: Operations Research, Universitas Indonesia
- ENIE606051: Decision, Uncertainty and Risk, Universitas Indonesia
- ENIE616027: Information System, Universitas Indonesia
- ENIE801006: Advanced Statistics for IE, Universitas Indonesia
- ENIE802537: Decision Analysis and Risk Management, Universitas Indonesia
- ENIE803433: Multivariate Data Analysis, Universitas Indonesia
- ENGE600003: Fundamentals of Physics, Universitas Indonesia
- ENGE600005: Statistics and Probability, Universitas Indonesia

Course	Position	Semester	Instructor Rating	Class Type
IEM 4723	GTA	Fall 2025	n.a.	In-person
IEM 5603	GTA	Summer 2025	n.a.	Online
IEM 5990	GTA	Fall 2024	n.a.	In-person
IEM 4913	GTA	Spring 2023	n.a.	In-person
IEM 4913	GTA	Fall 2022	n.a.	In-person
IEM 3813	GTA	Fall 2022	n.a.	In-person & Lab
IEM 4913	GTA	Spring 2022	n.a.	In-person
IEM 4913	GTA	Fall 2021	n.a.	In-person
IEM 3503	GTA	Summer 2021	n.a.	Online
IEM 3513	GTA	Spring 2021	n.a.	In-person

Course	Position	Semester	Instructor Rating	Class Type
ENGE600001	Instructor	Ganjil 2020	5.42 of 6.00	Regular
ENGE600001	Instructor	Ganjil 2019	5.18 of 6.00	Regular
ENGE600001	Instructor	Ganjil 2018	4.95 of 6.00	Regular
ENGE600002	Instructor	Genap 2020	5.20 of 6.00	Regular
ENGE600002	Instructor	SP 2019	5.53 of 6.00	Regular
ENGE600002	Instructor	Genap 2019	5.02 of 6.00	Regular
ENGE600002	Instructor	SP 2018	5.42 of 6.00	Regular
ENGE600011	Instructor	SP 2019	4.90 of 6.00	Regular
ENIE601002	Instructor	Ganjil	5.73 of	Regular

Course	Position	Semester	Instructor Rating	Class Type
		2020	6.00	
ENIE601002	Instructor	Ganjil 2019	5.59 of 6.00	Regular
ENIE601002	Instructor	Ganjil 2018	5.40 of 6.00	Regular
ENIE601004	Instructor	Ganjil 2020	5.70 of 6.00	Regular
ENIE603006	Instructor	Ganjil 2020	5.55 of 6.00	Regular
ENIE603006	Instructor	Ganjil 2019	5.09 of 6.00	International
ENIE603006	Instructor	Ganjil 2018	5.46 of 6.00	International
ENIE603006	Instructor	Ganjil 2017	5.04 of 6.00	International
ENIE603007	Instructor	Ganjil 2019	5.04 of 6.00	International
ENIE604011	Instructor	Genap 2019	5.59 of 6.00	International
ENIE604011	Instructor	Genap 2018	5.29 of 6.00	International
ENIE604015	Instructor	Genap 2020	4.90 of 6.00	Regular
ENIE604015	Instructor	Genap 2018	5.29 of 6.00	International
ENIE604016	Instructor	Genap 2020	5.16 of 6.00	International
ENIE604016	Instructor	Genap 2018	5.31 of 6.00	Regular

Course	Position	Semester	Instructor Rating	Class Type
ENIE606051	Instructor	Genap 2020	5.46 of 6.00	Regular
ENIE606051	Instructor	Genap 2019	5.59 of 6.00	International
ENIE606051	Instructor	Genap 2019	5.57 of 6.00	Regular
ENIE616027	Instructor	Genap 2020	5.56 of 6.00	International
ENIE616027	Instructor	Genap 2019	5.34 of 6.00	International
ENIE601002	GTA	Ganjil 2016	n.a.	Regular
ENIE603006	GTA	Ganjil 2016	n.a.	Regular
ENIE801006	GTA	Genap 2017	n.a.	Regular
ENIE802537	GTA	Genap 2017	n.a.	Regular
ENIE803433	GTA	Ganjil 2016	n.a.	Regular
ENGE600002	UTA	Genap 2015	n.a.	Regular
ENGE600003	UTA	Ganjil 2014	n.a.	Regular
ENGE600005	UTA	Genap 2014	n.a.	Regular

Notes: *GTA = Graduate Teaching Assistant* *UTA = Undergraduate Teaching Assistant*
Ganjil = similar to Fall Term *Genap = similar to Spring Term* *SP = similar to Summer Term*
Regular = class taught entirely in Indonesian *International = class taught entirely in English*

Services

Accreditation Team | Multiple Journal Reviewer | 39 Theses Examined

University Level Service

- **Staff Advisory Council, Oklahoma State University** - Representative (2024-2025)
  Represented the student perspective in university-level advisory discussions and institutional initiatives.
- **Strategic Initiative Provost Council, Oklahoma State University** - Member (2024)
  Participated in university strategic planning and initiatives under the Office of the Provost.
- **Capitol Cowboys Day, Oklahoma State University** - OSU Representative (2024, 2026)
  Selected by Oklahoma State University to represent the university at the Oklahoma State Capitol, engaging with state legislators alongside university leadership, alumni, and fellow student representatives.
- **Industrial Engineering Accreditation Team, Universitas Indonesia** - Member (2018, 2019, 2020)
  Contributed to accreditation preparation and evaluation activities related to Indonesian Industrial Engineering accreditation standards.
- **Industrial Engineering Curriculum Design Team, Universitas Indonesia** - Member (2018, 2019, 2020)
  Contributed to the review and development of the Industrial Engineering curriculum, including alignment of academic content and program objectives with educational and professional requirements.
- **Alumni Tracer Study and Department Development Team, Department of Industrial Engineering, Universitas Indonesia** - Lead (2018, 2019)
  Supported alumni tracer studies and analysis of graduate outcomes to inform departmental evaluation, academic planning, and program development.
- **Student Organization Mentor, Universitas Indonesia** - Faculty/Academic Mentor (2018-2020)
  Mentored student organizations and provided guidance on organizational activities, leadership development, and student initiatives.

Academic Reviewer

- European Journal of Operational Research
- Decision Analytics Journal
- Engineering Applications of Artificial Intelligence
- Journal of the American Medical Informatics Association
- Journal of Intelligent Manufacturing
- Social Sciences & Humanities Open Journal
- Sustainable Future Journal
- PeerJ Computer Science Journal
- Ain Shams Engineering Journal
- Indonesian Journal of Electrical Engineering and Computer Science

Conference Committee

- **Technical Program Chair**, Asia Pacific Conference on Research in Industrial and Systems Engineering (2018, 2019, 2020)
- **Scientific Committee**, Asia Pacific Conference on Research in Industrial and Systems Engineering (2020, 2021)

Thesis Examiner

Healthcare Analytics

ET039. Klairine Mariana Rustan, Improved Design and Construction of Hospital Blood Bank Inventory Management Using an Information System Through the Internet of Things Application, 2020

ET038. Zara Jesica Azra, Designing Radiological Health Equipment Maintenance Process Improvement with the Support of Internet of Things (IoT), 2020

ET035. Derryl Reyhan Zaradi, Analysis of Technology Model Adoption on Health Observation Through Smartwatches and Fitness Bands in Generation Y, X, and Baby Boomers in Indonesia, 2019

ET034. Wafa Makhasha Ulfah, Model Development to Determine Optimal Drug Inventory by Using Mixed Integer Linear Programming (MILP) in Public Health Services, 2019

ET017. Firas Ammar Akbar, Prioritization of User's Preferences for Using mHealth Services in Jakarta with Analytic Hierarchy Process (AHP) Approach, 2018

Behavioral Analytics and Text Mining

ET033. Nadya Syabila, Strategy Recommendation for Paylater Acceptance and Loyalty on Generation Z in Jabodetabek, 2020

ET032. Pelangi Putri, Designing Strategy Based on Experiential Values Factors Towards Customers' Revisit Intention in Coffee Shop Inside Universitas Indonesia, 2020

ET031. Doni Pratama, Analysis of the Effect of Learning Media (Digital Based Vs Print Based) on the Ability to Understand Material in Terms of Attention and Mental Effort, 2020

ET030. Chandatama Olivia Pramesti, Strategic Planning to Increase Intention to Donate Money on Indonesia's Online Crowdfunding Platform, 2020

ET029. Faishal Ahmad Andya Aji, Strategy Design to Increase Customer Loyalty of Fitness Center Based on the Factors of Perceived Quality and Service Convenience, 2020

ET028. Aliffa Safitri, Policy Formulation to Increase Market Share of Indonesian Islamic Banking Using System Dynamics Approach, 2020

ET024. Niko Muhammad Iskandar, Strategies to Increase Willingness to Lend Toward Peer-to-Peer Lending Companies Operated in Indonesia, 2019

ET023. Sri Wulandari Ningrum, Strategies to Increase Customer Satisfaction and Customer Loyalty Based on Customer Perceived Values Factors in Grandkemang Hotel Jakarta, 2019

ET022. Muhammad Zeid Masyhur, Designing Strategies to Improve Performance by Considering the Aspects of Sustainability in Clothing Services SME Using Fuzzy AHP Method, 2019

ET018. Annisa Nibras Adiba, Measurement of Customers and Employees Engagement Value with Human Sigma Method for Planning Company's Management Strategy: Case Study: HIS Wedding Event Planner, 2018

Optimization and Operations Management

ET037. Jihad Alif, Sustainable Production Modelling in Cement Industry Using System Dynamics, 2020

ET036. Andini Eka Ramadhani, Analysis of Drivers and Barriers to the Implementation of Integrated Plastic Waste Management in Indonesia Using DEMATEL, 2020

ET035. Dicky Nurachman, Feasibility Investment Analysis of Geothermal Project Development Using Financial Modelling (Study Case: Lumut Balai Geothermal Power Plant), 2020

ET034. Muhammad Shaddam Muzaki, Policy Analysis of the Geothermal Power Plant Capacity Development in Indonesia with System Dynamics Modeling, 2020

ET033. Bramanda Dwi Putra, Selection of Charging Station Technology to Support the Adoption of Electric Vehicles in Indonesia with the AHP-TOPSIS Method, 2020

ET027. Renaldhi Dwidinda Suharno, Quality Improvement of Lamination Process Results in the Packaging Industry by Using the Six Sigma Method, 2019

ET026. Galih Prier Aditiya, Remanufacturing Capacity Planning Model for Multigeneration Television Products in Closed Loop Supply Chain with System Dynamics Method, 2019

ET025. Sarah Rotua Harianja, Designing Strategies to Improve the Service Quality of Bus Rapid Transit (BRT) Transjakarta Corridor 13 Using Integration of Kano-Quality Function Deployment Method, 2019

ET016. Arin Wulandari, Comparative Analysis of Empennage Design Indonesian Aircraft Project Scheduling with Critical Path Method CPM and Critical Chain Project Management CCPM, 2018

ET015. Muhammad Rifai, Designing Scheduling Arrival of Container Trucks with Trucks Turnaround Time Method at Terminal Peti Kemas Selatan PT Pelabuhan Tanjung Priok, 2018

ET014. Irena Yasmin Djohan, Helicopter Flight Services Pricing for Tourism Purposes and Intercity Travel by Using Cost Plus Pricing Method, 2018

ET013. Agustina Windaryanti, Improvement Design of Supply Chain Management Implementation in Food MSME, 2018

ET012. Juan Siva, Design Improvement of SME's Business Process in Fashion Industry for Adopting E-Commerce Efficiently, 2018

ET011. Rossy Nicoline Subrana, Designing Strategies for Improving the Quality of Ragunan Wildlife Park Service Using Quality Deployment Method, 2018

ET010. Salma Tarizka Noor, Designing Strategies to Reduce Industrial Greenhouse Gas Emissions in Indonesia Using a System Dynamic Approach, 2018

ET009. Julia Eka Citra, Optimization of Water Truck Usage in Watering Coal Mine Using Integer Linear Programming, 2018

ET008. Fairuz Qalbi Andara, Risk Mitigation on Supply Chain of Livestock Industry Using House of Risk Approach, 2018

ET007. Vadhel Iqbal, Business Process and Card Procurement Mapping of PT MRT Jakarta at the Station Using Inventory Management Analysis, 2018

ET006. Ni Putu Dirda Putri, Optimization of Production Scheduling in Shoe Companies with Goal Programming Method, 2018

ET005. Faisal Ahmad Ridho, Investment Feasibility Study of N-219 Aircraft Purchasing for Pioneer Flight in North Kalimantan Province with Capital Budgeting Method, 2018

ET004. Rizky Anugerah Pratama, Improvement Recommendation to Eliminate Waste on the Production Process of Line Laminating Door Component of Joint Core Laminating Skin with Value Stream Mapping and Waste Assessment Model Method, 2018

ET003. Aldeina Putriandita, Designing Implementation Strategy for Internet of Things (IoT) on Logistic Transportation Sector in Indonesia Using Analytic Network Process Method, 2018

ET002. Panji Wara Rahmadhani, Lean Supply Chain Designing in SMEs Fashion E-Commerce Business Process Reengineering Approach, 2018

ET001. Fitri K. Tambunan, Repair Line Production with Lean Production System Based Approach VSM (Value Stream Mapping), 2018

ET0xx = i-th examined thesis

Additional Endorsement

"I believe Mr. Laoh is well-positioned to continue advancing this work. His background in industrial engineering, predictive analytics, and data-driven decision systems provides a strong foundation for work at the intersection of modeling methodology and real-world implementation. He appears to have developed a credible path toward contributions that are relevant not only to technical model development but also to the responsible use of AI in important application settings."

Mei Liu, Ph.D., Director of Predictive Analytics of Health Outcomes & Biomedical Informatics, University of Florida

"Mr. Laoh is well-prepared to advance this field. His background in industrial engineering, predictive modeling, analytics, and applied decision systems provides a solid foundation for connecting methodological advances with practical operational needs. Based on my review, he has already established a promising trajectory at the intersection of modeling, system performance, and decision support."

Xueping Li, Ph.D., Director of the Ideation Laboratory, University of Tennessee, Knoxville

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  [PhD, University of Manitoba 1988]
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